Phylogenetic Analysis Using NCBI and Galaxy

Group 2 - Sharks

BIO 300 (Cell & Molecular Biology)

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Introduction

Mitochondrial cytochrome c oxidase subunit I (COI) is a widely used genetic marker for studying the evolutionary relationships among animals due to its relatively slow rate of mutation. In this study, several shark species were selected because they represent an ecologically important group of marine organisms with diverse evolutionary histories. By comparing their mitochondrial COI DNA sequences, the evolutionary relationships among the selected shark species were inferred and compared with those reported in previous scientific studies.

Table 1. *Shark species, NCBI accession numbers, COI sequence lengths, and outgroup designation.*

Scientific name	Common name	NCBI accession	Length (bp)	Focal taxon / Outgroup
Sphyrna lewini	Hammerhead shark	MT456041.1	655 bp	Focal taxon
Negaprion acutidens	Lemon shark	GU674232.1	652 bp	Focal taxon
Ginglymostoma cirratum	Nurse shark	PQ347194.1	655 bp	Focal taxon
Galeocerdo cuvier	Tiger shark	FJ519959.1	652 bp	Focal taxon
Prionace glauca	Blue shark	HM422273.1	648 bp	Focal taxon
Rhincodon typus	Whale shark	MN759764.1	666 bp	Focal taxon
Carcharodon carcharias	Great white shark	KM212005.1	652 bp	Focal taxon
Alopias pelagicus	Thresher shark	KP193429.1	652 bp	Focal taxon
Mobula birostris	Oceanic manta ray	GU673825.1	652 bp	Outgroup

Methods

The mitochondrial COI sequences of eight shark species and one outgroup (*Mobula birostris*) were downloaded from the NCBI database. The sequences were uploaded to Galaxy and combined into a single dataset. Multiple sequence alignment was performed using MAFFT to compare similarities and differences among the sequences. A consensus sequence was then generated from the alignment to identify conserved and variable sites. Finally, a phylogenetic tree was constructed using FastTree, with *Mobula birostris* serving as the outgroup to infer the evolutionary relationships among the selected shark species.

The following images are the main steps in making a phylogenetic tree using the FASTA file in NCBI and running it on Galaxy.

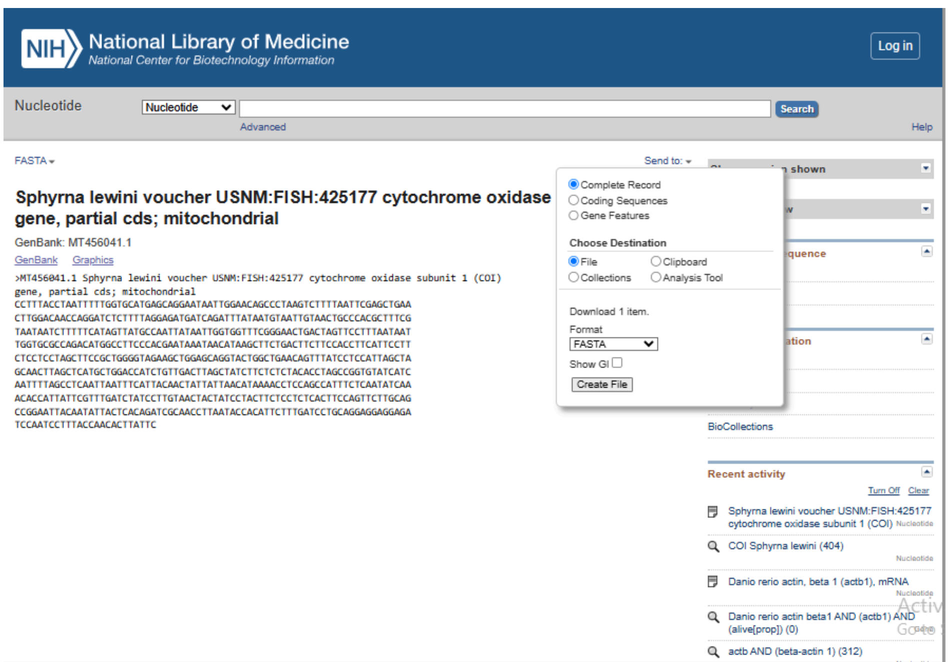

Figure 1. *Download the FASTA file from NCBI. Begin by retrieving your sequence of interest (e.g., COI gene) from the NCBI Nucleotide database and saving it in FASTA format. Change its filename to the organism you've chosen. This file will serve as the input for the next steps in Galaxy.*

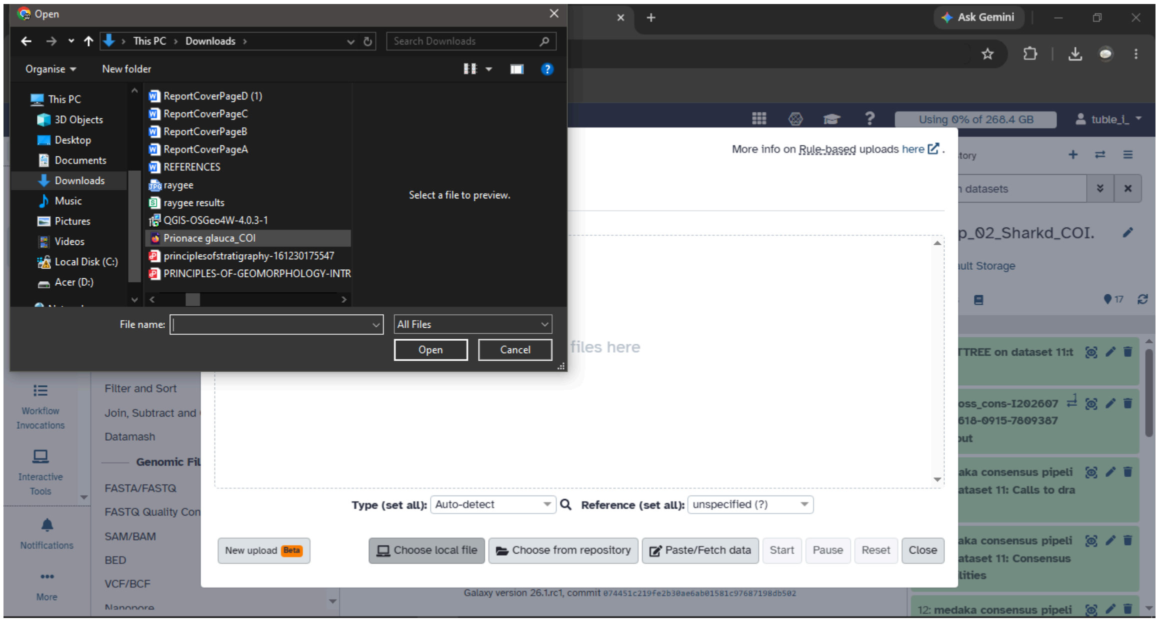

Figure 2. *Upload the FASTA files to Galaxy. On the left panel, select Upload, choose Local File, and then locate the FASTA file you previously saved from NCBI. This will import your sequence into Galaxy for alignment and tree construction.*

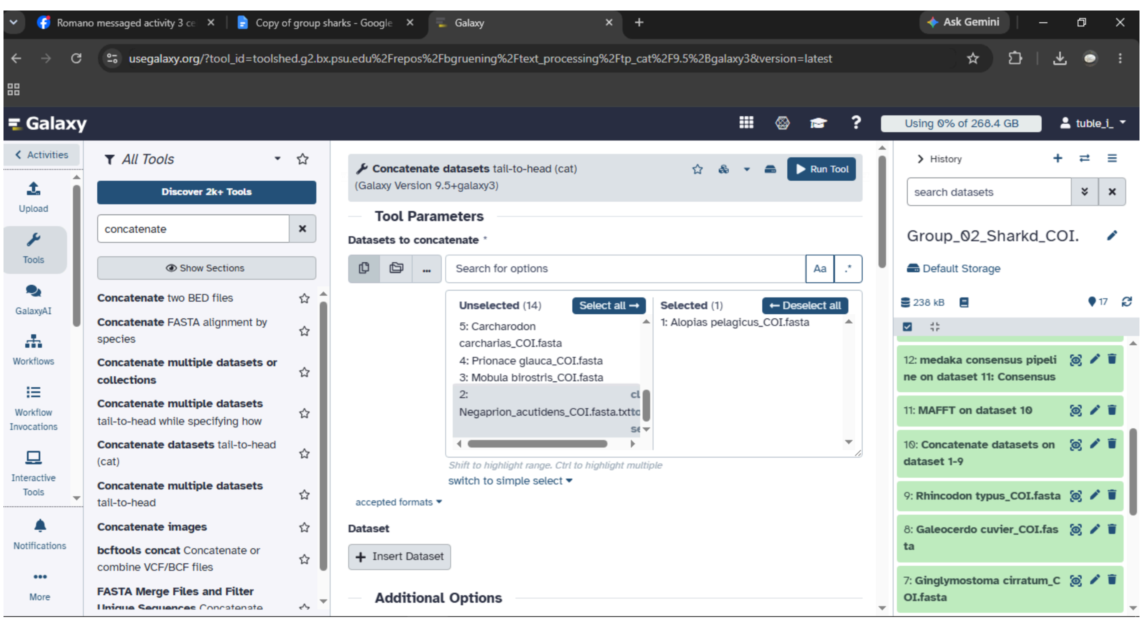

Figure 3. *Convert to MultiFASTA. Go to the Tools section in Galaxy and select the option that allows you to combine or concatenate sequences. This will turn your single FASTA file into a multiFASTA file, which is required for multiple sequence alignment and phylogenetic tree construction.*

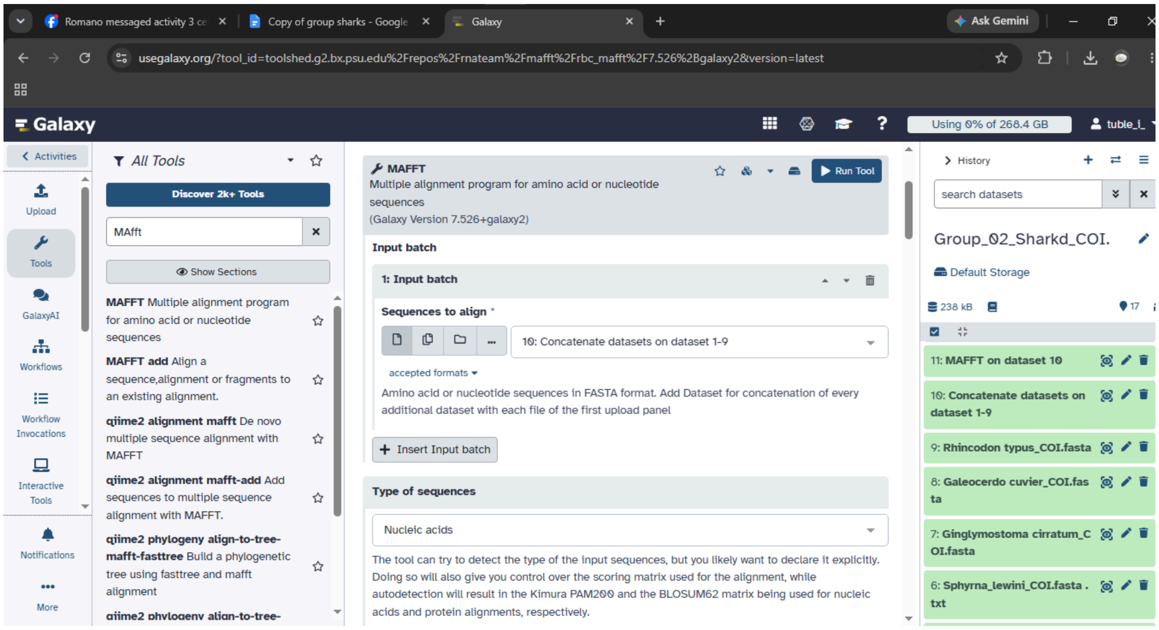

Figure 4. *Run MAFFT Alignment. In the Tools panel, search for MAFFT and select it for sequence alignment. Then choose the concatenate database option and apply it to datasets 1–9 to prepare your sequences for phylogenetic analysis.*

	Id	Actual Id	Begin	End	Link	Real Length	Aligned Leng...	Left Gap
1	KM212005.1	KM212005.1	1	652		652	652	2
2	HM422273.1	HM422273.1	1	648		648	648	2
3	GU673825.1	GU673825.1	1	652		652	652	2
4	GU674232.1	GU674232.1	1	652		652	652	2
5	MN759764.1	MN759764.1	1	666		666	666	0
6	FJ519959.1	FJ519959.1	1	652		652	652	2
7	PQ347194.1	PQ347194.1	1	655		655	655	2
8	MT456041.1	MT456041.1	1	655		655	655	2
9	KP193429.1	KP193429.1	1	652		652	652	2

Figure 5. *Visualize MAFFT Alignment In the Tools panel, open MAFFT visualization and select Multiple Sequence Alignment along with Sequence Logo. This allows you to clearly see the alignment, highlighting sequence differences and similarities across your dataset.*

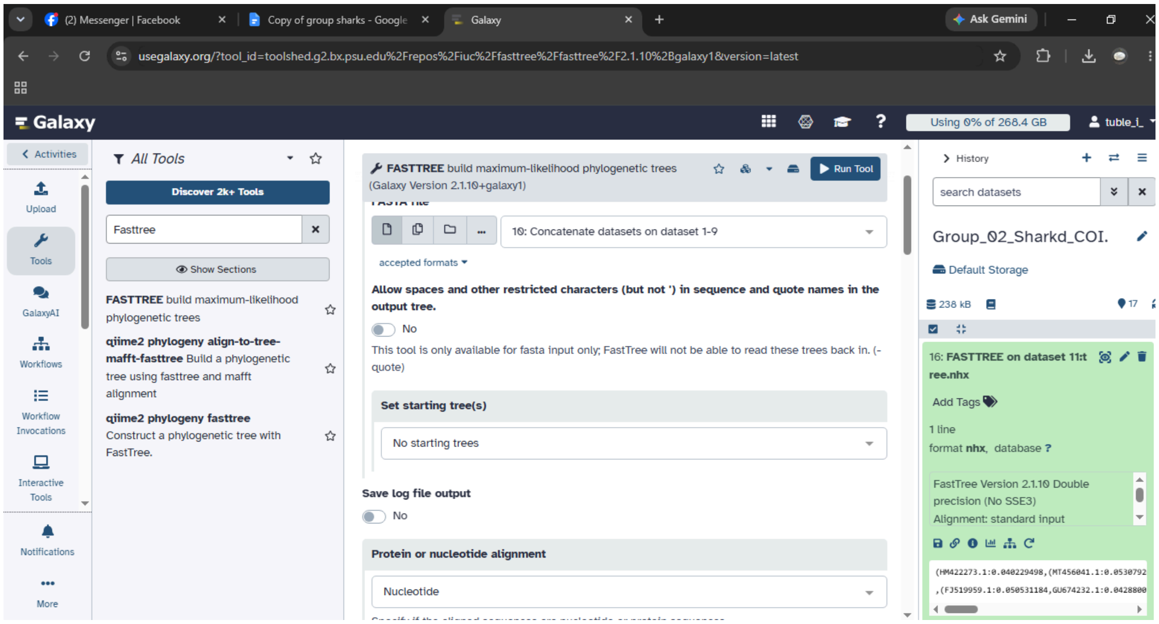

Figure 6. *Build a Tree with FastTree. In the Tools panel, search for FastTree and select it to construct a phylogenetic tree using the maximum-likelihood method. This will generate the evolutionary relationships based on your aligned sequences.*

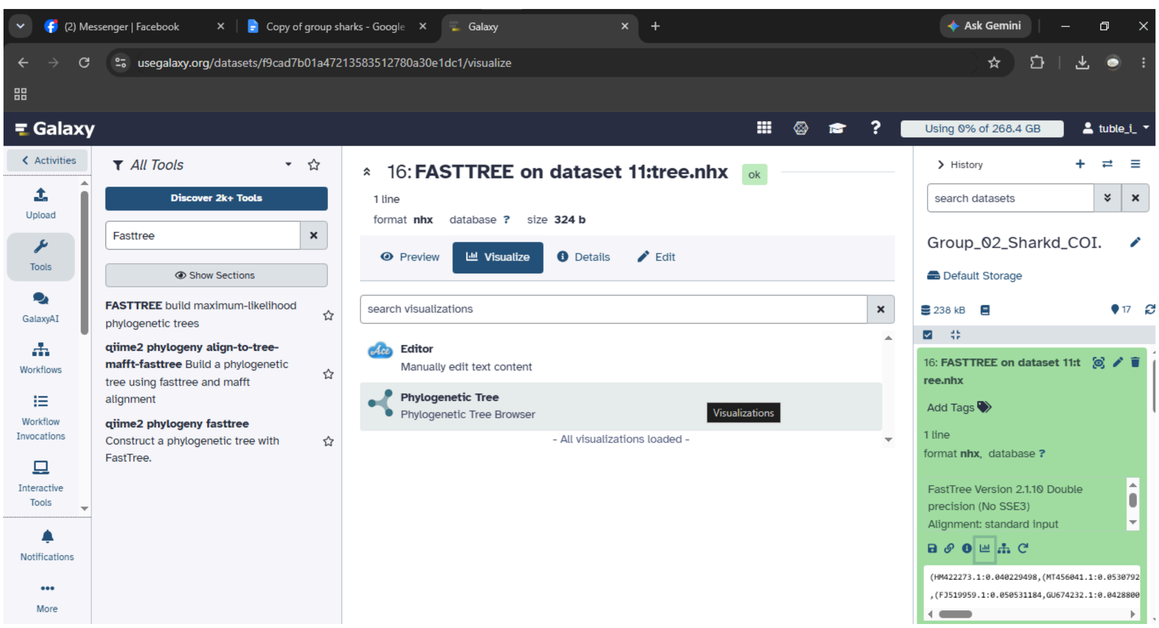

Figure 7. *Visualize FastTree Output In the Tools panel, open the FastTree visualization option to display the phylogenetic tree. This allows you to see the branching patterns and evolutionary relationships among your aligned sequences.*

Summary of Conserved and Variable Sites

The consensus sequence was generated from the aligned COI sequences of the eight shark species and one outgroup. Most nucleotide positions were conserved, meaning they were the same across the species, indicating that the COI gene is highly conserved among the selected species. However, some positions were variable, where the nucleotides differed between species. These variable sites provide useful information for distinguishing species and were used to infer the evolutionary relationships shown in the phylogenetic tree.

The phylogenetic tree grouped the selected shark species into several clades based on their mitochondrial COI sequences. The Nurse shark (*Ginglymostoma cirratum*) and Whale shark (*Rhincodon typus*) formed a pair of sister taxa, while the Great white shark (*Carcharodon carcharias*) and Thresher shark (*Alopias pelagicus*) also appeared closely related, indicating that each pair shares a more recent common ancestor. The Oceanic manta ray (*Mobula birostris*) was designated as the outgroup for this analysis. Although it was expected to branch outside the shark taxa, it grouped closer to some shark lineages in the generated tree. This unexpected placement may be due to the use of a single mitochondrial COI gene, limited taxon sampling, sequence quality, or methodological factors. The branch lengths represent the amount of genetic difference between species, with shorter branches indicating closer relationships and longer branches indicating greater genetic divergence. Since this analysis is based on a single mitochondrial gene, it should not be considered the definitive species phylogeny, as analyses using multiple genes or whole genomes can provide a more complete picture of evolutionary relationships.

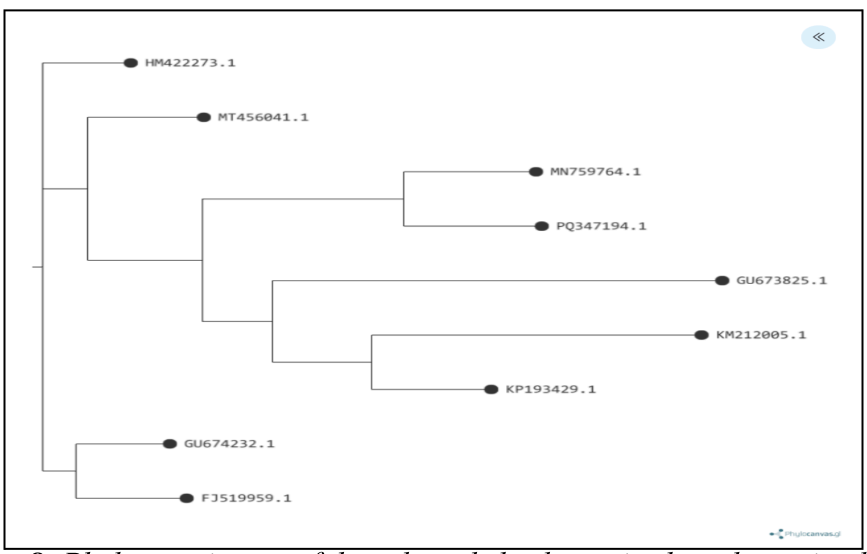

Figure 8. *Phylogenetic tree of the selected shark species based on mitochondrial COI sequences generated using FastTree.*

Comparison with Published Studies

The phylogenetic tree showed that the Nurse shark (*Ginglymostoma cirratum*) and the Whale shark (*Rhincodon typus*)formed a closely related group, while the Great white shark (*Carcharodon carcharias*) and the Thresher shark (*Alopias pelagicus*) also appeared closely related. These results are generally consistent with the findings of Naylor et al. (2012), who reported evolutionary relationships among sharks using molecular sequence data. Similar to their findings, the phylogenetic tree generated in this study grouped several shark species into closely related clades. However, the placement of the outgroup differed, likely because this study analyzed only the mitochondrial COI gene and included fewer taxa. However, our tree differed in the placement of some species, likely because only the mitochondrial COI gene and a small number of taxa were analyzed.

In addition, Ward et al. (2005) reported that mitochondrial COI sequences are effective for distinguishing fish species through DNA barcoding, supporting the use of COI in our study for identifying shark species. However, they also noted that COI is primarily a species identification marker and may not fully resolve deeper evolutionary relationships when used alone. Therefore, the generated phylogenetic tree should be interpreted with caution and ideally supported by analyses using additional genes. *(This statement is based on the widely cited Ward et al. DNA barcoding study.)*

REFERENCES:

Naylor, G. J. P., Caira, J. N., Jensen, K., Rosana, K. A. M., White, W. T., & Last, P. R. (2012). *A DNA sequence-based approach to the identification of shark and ray species and its implications for global elasmobranch diversity and parasitology*. *Bulletin of the American Museum of Natural History, 367*, 1–262. https://doi.org/10.1206/754.1

Ward, R. D., Zemlak, T. S., Innes, B. H., Last, P. R., & Hebert, P. D. N. (2005). *DNA barcoding Australia's fish species*. *Philosophical Transactions of the Royal Society B: Biological Sciences, 360*(1462), 1847–1857. https://doi.org/10.1098/rstb.2005.1716